

Laundry Management System

Problem Statement & Queries Explanation

1. Problem Statement

Context of Use

“Kigali Fresh Laundry,” a busy neighbourhood laundry shop in Kigali, currently records customer orders, service prices, and payments by hand in notebooks. This causes lost order slips, incorrect billing, no visibility into which attendant is handling which order, and no historical record for decision-making.

Target Users

- Shop Manager — oversees operations and reports.
- Front-Desk Officer — registers customer orders and payments.
- Laundry Attendants — process, wash, dry, and prepare customer items.
- Customers — receive accurate, transparent, and timely laundry services.

Project Objectives

- Centralize customer, order, payment, and service data in Oracle Database.
- Track every order from drop-off through washing, ironing, and pickup.
- Automate pricing and billing calculations to minimize human error.
- Enforce data integrity using constraints, keys, and secure database access.
- Provide an audit trail of important database changes for accountability.

Expected Benefits

- Faster and more accurate order processing.
- Real-time tracking of laundry order status.
- Accurate and transparent billing for customers.
- Reliable records for reporting, audits, and performance analysis.
- Reduced fraud and improved accountability through activity logging.

2. SQL Queries Explanation

The following section explains the purpose of the SQL and PL/SQL scripts included in the project repository.

DDL (CREATE TABLE)

Creates all database tables such as Customers, Orders, Order_Items, Services, Payments, Employees, Audit_Log, and related structures. Primary keys, foreign keys, constraints, and relationships enforce data integrity.

DML (INSERT, UPDATE, DELETE)

Adds sample records, updates existing information when changes occur, and removes unnecessary records while maintaining database consistency.

SELECT Queries

Retrieve information such as customer details, pending orders, completed orders, payments, and daily operations for staff.

JOIN Queries

Combine data from multiple tables to display complete order information, including customer, employee, services, and payment details.

Aggregate Queries

Use COUNT, SUM, AVG, MIN, and MAX to generate reports such as total revenue, number of orders, and average service cost.

GROUP BY and HAVING

Group records by categories such as service type or employee and filter summarized results for reporting.

Subqueries

Retrieve records based on the results of another query, such as customers with unpaid orders or high-value transactions.

Views

Create reusable virtual tables that simplify reporting and improve security by exposing only required information.

Sequences

Automatically generate unique IDs for records such as customers and orders.

Indexes

Improve query performance by speeding up searches on frequently accessed columns.

PL/SQL Procedures

Automate repetitive business operations such as updating order status or processing payments.

PL/SQL Functions

Return calculated values such as total order cost or outstanding balance.

Triggers

Automatically record changes in the Audit_Log table and enforce business rules whenever INSERT, UPDATE, or DELETE operations occur.

CTEs (Common Table Expressions)

Simplify complex SQL statements by breaking them into readable temporary result sets before the final query executes.

Window Functions

Perform advanced analysis such as ranking customers by spending, calculating running totals, and comparing employee performance without collapsing rows.

Conclusion

This Oracle-based Laundry Management System replaces manual record keeping with a secure, centralized database. The included SQL and PL/SQL scripts demonstrate database design, transaction processing, reporting, automation, auditing, and analytical capabilities suitable for academic and practical use.